

DEEPAK C S

AI & Cloud Data Engineer • 4+ Years Experience • EY (Ernst & Young)

sureshdeepak556@gmail.com | +91 8110096419 | [LinkedIn](#) | [GitHub](#) | Tamil Nadu, India

PROFESSIONAL SUMMARY

Results-driven AI & Cloud Data Engineer with 4.4 years of enterprise experience architecting modern Lakehouse platforms and embedding AI automation into mission-critical data workflows. At EY, leading a multi-source migration to a Databricks–Snowflake Lakehouse while building AI-powered multi-agent systems that compress 1–2-week UAT cycles to near-zero human effort. Consistent record of **0 production failures**, **100% on-time delivery**, and measurable pipeline accuracy gains (90–95%+) across Azure, Databricks, and Snowflake ecosystems. Adept at critical edge-case thinking that prevents issues before they reach production.

WORK EXPERIENCE

AI & Cloud Data Engineer | EY — Ernst & Young

Apr 2025 – Present

- Architecting a modern multi-layer **Lakehouse**: ingesting from **Oracle**, **DB2**, **Avro**, **REST API**, and **Parquet** into **Databricks External Delta Tables** (Stage → Analytics on **ADLS Gen2**) — the unified enterprise data foundation.
- Engineered zero-copy **Snowflake External Iceberg Tables** mapped to the same **ADLS** storage as Databricks, eliminating redundant data movement and enabling seamless cross-platform analytics without duplication.
- Built end-to-end **DIM & FACT** data models in **Snowflake** via **Matillion ETL**; orchestrated **Databricks–Matillion** job dependencies end-to-end using **Autosys** for reliable, automated pipeline execution.
- Designed and deployed an **AI-powered multi-agent UAT automation system (MCP + JIRA integration)**: autonomously reads pipeline context, generates **SQL test cases** from business requirements, executes on **Databricks & Snowflake**, validates results, and produces an audit report — compressing a **1–2-week manual validation cycle** to near-zero effort, significantly reducing business cost.
- Delivered targeted **query & pipeline optimizations** across massive datasets, reducing critical job runtimes through systematic **edge-case analysis** and proactive performance modelling.
- 0 production failures** and **100% on-time delivery** across all migration workstreams — driven by rigorous pre-production validation and proactive risk mitigation.

Technology Analyst | Infosys

Oct 2024 – Apr 2025

- Managed complex enterprise job scheduling in **Control-M**, ensuring precise dependency chains and uninterrupted execution across production pipelines.
- Led migration of legacy **Azure** jobs into a standardized **Python/Pyspark** framework; achieved **30% reduction in error rates** through systematic test coverage and root-cause-driven bug resolution.

Senior Systems Engineer | Infosys

Feb 2022 – Oct 2024

- Spearheaded **Mainframe** and **Informatica Legacy ETL** migration to **Azure Cloud ETL** using custom data ingestion frameworks, accelerating enterprise cloud modernization at scale.
- Engineered **incremental (watermark-based)** and **full-reload pipelines** from **Teradata**, **SAP HANA**, **DB2**, and **Oracle** into **Databricks Delta tables** — achieving **90%+ accuracy** and efficiency.
- Designed an automated post-ingestion **data validation framework** between **Azure** and **Teradata**, automating **90% of QA workload** and significantly compressing validation cycle time.
- Built an **ADLS Parquet vs. Teradata** comparison pipeline delivering **95% data precision**, enabling confident legacy system decommissioning.

Systems Engineer | Infosys

Dec 2021 – Feb 2022

- Analysed critical **Informatica** workflows, delivering a **50% improvement** in system reliability and pipeline stability.
- Streamlined reporting processes, improving team productivity and client satisfaction scores by **20%**.

KEY PROJECTS

AI Multi-Agent UAT Automation | EY, 2025 | Stack: Multi-Agent AI, MCP, JIRA, Databricks, Snowflake, Python

- Architected an **AI multi-agent system** using **MCP (JIRA integration)** to fully automate **SQL-based UAT validation**. System reads pipeline context, auto-generates test SQL with expected outputs, executes across **Databricks & Snowflake**, cross-validates results, and produces a compliance audit report — eliminating **1–2 weeks of manual QA effort** per release cycle.

XML Medical Records Streaming Pipeline | Databricks Hackathon | [GitHub](#)

- Built real-time **XML ingestion pipeline** using **Spark Structured Streaming & Autoloader**, achieving **98% near real-time accuracy**; delivered live dashboards via **Databricks SQL Warehouse** with **95% precision** and automated saturation alerts.

Delta Table Validation Automation | [GitHub](#)

- Engineered a **Pyspark** automation framework for **DEV/PROD Delta table** validation using minus queries and **parallel cluster execution**; ensured cross-environment data consistency with zero manual intervention.

Weather Analytics Platform — Microsoft Fabric | [GitHub](#)

- End-to-end real-time analytics using **Azure Data Factory, Spark Structured Streaming, Pyspark, SQL**, and **Power BI** on **Microsoft Fabric** for live data modelling and dashboard delivery.

CERTIFICATIONS

- Databricks Certified Data Engineer Professional ([Active](#))
- Snowflake SnowPro Core Certified ([Active](#))
- Microsoft Certified: Fabric Analytics Engineer – DP-600 ([Active](#))
- GitHub Copilot Certification – GH-300 ([Active](#))
- Microsoft Certified: Azure Data Fundamentals – DP-900 ([Active](#))
- Microsoft Certified: Azure Fundamentals – AZ-900 ([Active](#))

TECHNICAL SKILLS

AI & Agents	Multi-Agent Systems, MCP Protocol, GitHub Copilot, AI Automation
Languages	Python, Pyspark, SQL
Cloud & Platforms	Databricks, Snowflake, Azure Data Factory, ADLS Gen2, Microsoft Fabric
Big Data	Apache Spark, Spark Structured Streaming, Delta Lake, Apache Iceberg, Autoloader, Hadoop
ETL & Orchestration	Matillion ETL, Autosys, Control-M, ETL/ELT Pipelines, Informatica
Databases	Teradata, Oracle, DB2, SAP HANA, Microsoft SQL Server
Data Formats	Delta Tables, Apache Iceberg, Parquet, Avro, REST API, XML

EDUCATION

Bachelor of Engineering — Mechanical

2016 – 2020

Panimalar Engineering College, Anna University | **CGPA: 8.55 / 10.0**